

Surfaces. Spherical and cylindrical coordinates.

[SUBMIT ANSWERS TO ALL QUESTIONS]

A surface can be defined implicitly by an equation of the form $f(x, y, z) = C$, or given in parametric form: $x = x(u, v)$, $y = y(u, v)$, $z = z(u, v)$ [i.e., $\mathbf{r} = \mathbf{r}(u, v)$].

A vector $\mathbf{n}$ normal to the surface can be found as ∇f or $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$.

The equation of the *tangent plane* to a surface at point $\mathbf{r}_0$ is $(\mathbf{r} - \mathbf{r}_0) \cdot \mathbf{n} = 0$.

Spherical coordinates: radius r , polar angle θ , azimuth ϕ ($0 \leq r < \infty$, $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$),

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta,$$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \cos \theta = z / \sqrt{x^2 + y^2 + z^2}, \quad \tan \phi = y/x.$$

Length element (squared), spherical surface element and volume element:

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2, \quad dS = r^2 \sin \theta d\theta d\phi, \quad dV = r^2 \sin \theta dr d\theta d\phi.$$

Cylindrical coordinates: radius ρ , angle ϕ , z ($0 \leq \rho < \infty$, $0 \leq \phi \leq 2\pi$, $-\infty < z < \infty$),

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z,$$

$$\rho = \sqrt{x^2 + y^2}, \quad \tan \phi = y/x, \quad z = z.$$

Length element (squared), cylindrical surface element and volume element:

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2, \quad dS = \rho d\phi dz, \quad dV = \rho d\rho d\phi dz.$$

1. A parabolic surface (paraboloid of revolution) is described by the equation $z = 1 - x^2 - y^2$.

(a) Verify that point $\mathbf{r}_0 = (1, 1, -1)$ lies on this surface. Write the equation of the surface in the form $f(x, y, z) = 1$ and find the normal $\mathbf{n} = \nabla f$ to the surface at point $\mathbf{r}_0$. Hence, find the equation of the tangent plane at this point. [15]

(b) Show that this surface can be described in parametric form by

$$\mathbf{r}(u, v) = u\mathbf{i} + v\mathbf{j} + (1 - u^2 - v^2)\mathbf{k}.$$

Verify that $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$ is the same vector as $\mathbf{n}$ in part (a) at any point on the surface. [15]

2. (a) Show that the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, can be parameterised as [10]

$$\mathbf{r} = a \sin u \cos v \mathbf{i} + b \sin u \sin v \mathbf{j} + c \cos u \mathbf{k} \quad (0 \leq u \leq \pi, 0 \leq v \leq 2\pi).$$

(b) Show that $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$ is collinear to $\mathbf{n} = \nabla f$, where $f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}$. [10]

3. For two points in Cartesian coordinates, $A(1, 2, 2)$ and $B(-1, -1, -2)$, find their cylindrical coordinates (ρ, ϕ, z) and spherical coordinates (r, θ, ϕ) . [20]

4. Using cylindrical coordinates, the position vector can be written as

$$\mathbf{r} = \rho \cos \phi \mathbf{i} + \rho \sin \phi \mathbf{j} + z \mathbf{k}.$$

Find vectors $(d\mathbf{r})_\rho = \frac{\partial \mathbf{r}}{\partial \rho} d\rho$, $(d\mathbf{r})_\phi = \frac{\partial \mathbf{r}}{\partial \phi} d\phi$ and $(d\mathbf{r})_z = \frac{\partial \mathbf{r}}{\partial z} dz$, and show that the volume of the parallelepiped formed by vectors $(d\mathbf{r})_\rho$, $(d\mathbf{r})_\phi$ and $(d\mathbf{r})_z$ is $dV = \rho d\rho d\phi dz$. [30]

[Hint: this volume can be found as $(d\mathbf{r})_\rho \cdot [(d\mathbf{r})_\phi \times (d\mathbf{r})_z]$.]

MT11021 Problem Sheet 21.

Surfaces, Spherical and Cylindrical Co-ordinates: Solutions.

Q1.

(a). The equation of the surface can be written as

$$x^2 + y^2 + z = 1 \quad \dots (*)$$

Substituting the co-ordinates of the point $r_0 = (1, 1, -1)$ into the left-hand side, we have

$$1^2 + 1^2 + (-1) = 2 - 1 = 1$$

which proves that the point r_0 lies on the surface.

Denoting the left hand side of (*) by

$$F(x, y, z) = x^2 + y^2 + z$$

we find the normal

$$\underline{n} = \nabla F = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) = (2x, 2y, 1)$$

$$\therefore \underline{n} = 2x \underline{i} + 2y \underline{j} + \underline{k} \quad \dots (**)$$

at $r_0 = (1, 1, -1)$

$$\underline{n} = 2 \underline{i} + 2 \underline{j} + \underline{k}$$

The equation of the tangent plane is

$$(\underline{r} - \underline{r}_0) \cdot \underline{n} = 0$$

$$\underline{r} \cdot \underline{n} = \underline{r}_0 \cdot \underline{n}$$

giving

$$(\underline{x} \underline{i} + \underline{y} \underline{j} + \underline{z} \underline{k}) \cdot (2\underline{i} + 2\underline{j} + \underline{k}) = (\underline{i} + \underline{j} - \underline{k}) \cdot (2\underline{i} + 2\underline{j} + \underline{k})$$

$$2x + 2y + z = 2 + 2 - 1$$

$$2x + 2y + z = 3$$

is the required equation of the tangent plane. [15]

(b) Let

$x = u$, $y = v$ then

$$z = 1 - x^2 - y^2 = 1 - u^2 - v^2$$

and we have the equation of the surface in parametric form.

$$\underline{r}(u, v) = u \underline{i} + v \underline{j} + (1 - u^2 - v^2) \underline{k}$$

$$\frac{\partial \underline{r}}{\partial u} = \underline{i} - 2u \underline{k}$$

$$\frac{\partial \underline{r}}{\partial v} = \underline{j} - 2v \underline{k}$$

$$\frac{\partial \underline{r}}{\partial u} \times \frac{\partial \underline{r}}{\partial v} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 0 & -2u \\ 0 & 1 & -2v \end{vmatrix}$$

$$= \underline{i} \begin{vmatrix} 0 & -2u \\ 1 & -2v \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & -2u \\ 0 & -2v \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$= \underline{i} (2u) - \underline{j} (-2v) + \underline{k}$$

$$= 2u \underline{i} + 2v \underline{j} + \underline{k}$$

Since $u = x$ and $v = y$

$$\frac{\partial \underline{r}}{\partial u} \times \frac{\partial \underline{r}}{\partial v} = 2x \underline{i} + 2y \underline{j} + \underline{k}$$

which is the same as $\underline{n}$ in (**)
[15].

(2) (a) From the parametric form

$$\underline{r} = a \sin u \cos v \underline{i} + b \sin u \sin v \underline{j} + c \cos u \underline{k}$$

we have

$$x = a \sin u \cos v, \quad y = b \sin u \sin v, \quad z = c \cos u.$$

Substituting these into the left hand side of

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \dots (*)$$

we have

$$\frac{a^2 \sin^2 u \cos^2 v}{a^2} + \frac{b^2 \sin^2 u \sin^2 v}{b^2} + \frac{c^2 \cos^2 u}{c^2}$$

$$= \sin^2 u (\cos^2 v + \sin^2 v) + \cos^2 u$$

$$= \sin^2 u + \cos^2 u$$

$$= 1$$

ie the right hand side of (*)

[10].

(b).

$$F(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}$$

$$\underline{n} = \underline{\nabla} F = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{2z}{c^2} \right)$$

From the parametric form

$$\frac{\partial \underline{r}}{\partial u} = a \cos u \cos v \underline{i} + b \cos u \sin v \underline{j} - c \sin u \underline{k}$$

$$\frac{\partial \underline{r}}{\partial v} = -a \sin u \sin v \underline{i} + b \sin u \cos v \underline{j}$$

so that

$$\frac{\partial \underline{r}}{\partial u} \times \frac{\partial \underline{r}}{\partial v} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a \cos u \cos v & b \cos u \sin v & -c \sin u \\ -a \sin u \sin v & b \sin u \cos v & 0 \end{vmatrix}$$

$$= \underline{i} (bc \sin^2 u \cos v) - \underline{j} (-ac \sin^2 u \sin v) + \underline{k} (abc \cos u \sin u \cos^2 v + abc \cos u \sin u \sin^2 v)$$

$$= \sin u \left[bc \sin u \cos v \underline{i} + ac \sin u \sin v \underline{j} + abc \cos u \underline{k} \right]$$

$$= \sin u \left(abc \frac{a \sin u \cos v}{a^2} \underline{i} + abc \frac{b \sin u \sin v}{b^2} \underline{j} + abc \frac{c \cos u}{c^2} \underline{k} \right)$$

$$= abc \sin u \left(\frac{a \sin u \cos v}{a^2} \underline{i} + \frac{b \sin u \sin v}{b^2} \underline{j} + \frac{c \cos u}{c^2} \underline{k} \right)$$

Comparing with the parametric form we see

$$z = -2$$

$$\tan \phi = \frac{-1}{-1} = \frac{\sin \phi}{\cos \phi} \quad \begin{matrix} < 0 \\ < 0 \end{matrix} \quad \text{3rd quadrant.}$$

$$\tan \phi = 1 \Rightarrow \phi = \pi/4 \quad \text{in 1st quadrant}$$

$$\phi = \pi/4 + \pi = 5\pi/4 \quad \text{in 3rd quadrant.}$$

$$(r, \phi, z) = (\sqrt{2}, 5\pi/4, -2) \quad [10].$$

Spherical: $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \cos \theta = \frac{z}{\sqrt{x^2 + y^2 + z^2}}, \quad \tan \phi = \frac{y}{x}.$$

Point A $x=1, y=2, z=2$.

$$r = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3.$$

radius

$$\cos \theta = \frac{2}{\sqrt{9}} = \frac{2}{3} \Rightarrow \theta = \arccos \frac{2}{3}$$

$$\approx 0.841 \text{ rad}$$

= polar angle.

$\tan \phi = \frac{y}{x} = 2$ and the angle is the same as in cylindrical co-ordinates

$$\phi = \arctan 2 \approx 1.107 \text{ rad}$$

$$\therefore (r, \theta, \phi) = (3, 0.841 \text{ rad}, 1.107 \text{ rad}).$$

$$\frac{\partial \underline{r}}{\partial u} \times \frac{\partial \underline{r}}{\partial v} = abc \sin u \left(\frac{x}{a^2} \underline{i} + \frac{y}{b^2} \underline{j} + \frac{z}{c^2} \underline{k} \right)$$

$$= \frac{abc \sin u}{2} \underline{n}$$

$\therefore \frac{\partial \underline{r}}{\partial u} \times \frac{\partial \underline{r}}{\partial v}$ and $\underline{n}$ are collinear (or parallel).

[10].

3. $A(1, 2, 2)$ $B(-1, -1, -2)$.

Cylindrical $x = \rho \cos \phi$, $y = \rho \sin \phi$, $z = z$.

$$\rho = \sqrt{x^2 + y^2} ; \tan \phi = \frac{y}{x} ; z = z.$$

Point A $x=1$, $y=2$, $z=2$.

$$\rho = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\tan \phi = \frac{2}{1} = \frac{\sin \phi}{\cos \phi} > 0$$

1st quadrant

$$\therefore \phi = \tan^{-1}(2) \text{ or } \arctan 2.$$

$$\phi \simeq 1.107 \text{ rads.}$$

The z -co-ordinate is unchanged $z=2$.

$$(\rho, \phi, z) = (\sqrt{5}, 1.107 \text{ rad}, 2).$$

Point B $x=-1$, $y=-1$, $z=-2$.

$$\rho = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

Point B $x = -1, y = -1, z = -2.$

radius $r = \sqrt{(-1)^2 + (-1)^2 + (-2)^2} = \sqrt{1+1+4}$
 $= \sqrt{6}.$

polar angle $\cos \theta = \frac{-2}{\sqrt{6}}.$

$\theta = \arccos\left(\frac{-2}{\sqrt{6}}\right) \approx 2.526 \text{ rad}$

azimuth angle ϕ is the same as in cylindrical co-ordinates $\phi = 5\pi/4$

$(r, \theta, \phi) = (\sqrt{6}, 2.526 \text{ rad}, 5\pi/4).$

[10].

Q4. In cylindrical co-ordinates

$$\underline{r} = \rho \cos \phi \underline{i} + \rho \sin \phi \underline{j} + z \underline{k}$$

$$(d\underline{r})_\rho = \frac{\partial \underline{r}}{\partial \rho} d\rho = (\cos \phi \underline{i} + \sin \phi \underline{j}) d\rho$$

$$\begin{aligned} (d\underline{r})_\phi &= \frac{\partial \underline{r}}{\partial \phi} d\phi = (-\rho \sin \phi \underline{i} + \rho \cos \phi \underline{j}) d\phi \\ &= \rho (-\sin \phi \underline{i} + \cos \phi \underline{j}) d\phi \end{aligned}$$

$$(d\underline{r})_z = \frac{\partial \underline{r}}{\partial z} dz = \underline{k} dz.$$

The volume of the parallelepiped formed by vectors $(d\underline{r})_\rho$, $(d\underline{r})_\phi$ and $(d\underline{r})_z$ is

$$dV = (dr)_\rho \cdot ((dr)_\phi \times (dr)_z)$$

remember

$$\underline{a} \cdot (\underline{b} \times \underline{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$$

$$\therefore dV = (dr)_\rho \cdot ((dr)_\phi \times (dr)_z)$$

$$= \begin{vmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{vmatrix} \underbrace{dp \, \rho \, d\phi \, dz}_{\text{numerical factors can be taken out}}$$

$$= (\cos \phi (\cos \phi - 0) - \sin \phi (-\sin \phi - 0) + 0) \cdot \rho \, dp \, d\phi \, dz$$

$$= (\cos^2 \phi + \sin^2 \phi) \rho \, dp \, d\phi \, dz$$

[30].

$$\therefore dV = \rho \, dp \, d\phi \, dz$$

which is the volume element in cylindrical co-ordinates. One can also show that $(dr)_\rho$, $(dr)_\phi$ and $(dr)_z$ are mutually orthogonal by taking the scalar product in pairs. This volume element dV can also be determined using the Jacobian for the transformation from Cartesian to cylindrical co-ordinates (see example on canvas).